Evolution of electron localization with Co^{2+} variations in $Co_xFe_{3-x}O_4$ hollow spheres for enhanced wave absorption

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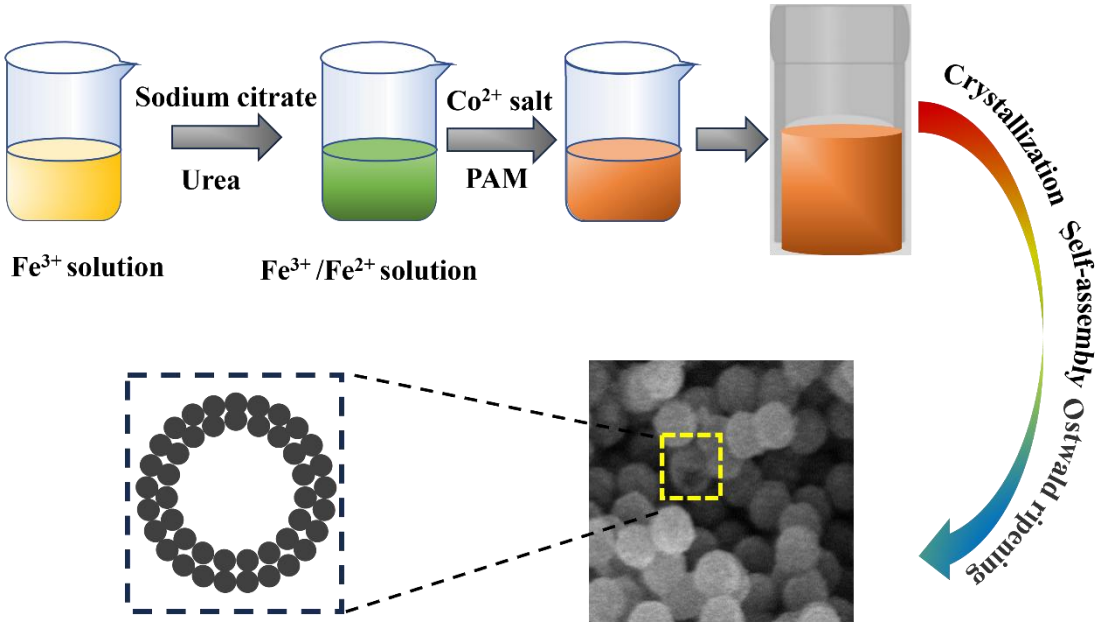


Figure S1. Hollow $Co_xFe_{3-x}O_4$ samples of the products.

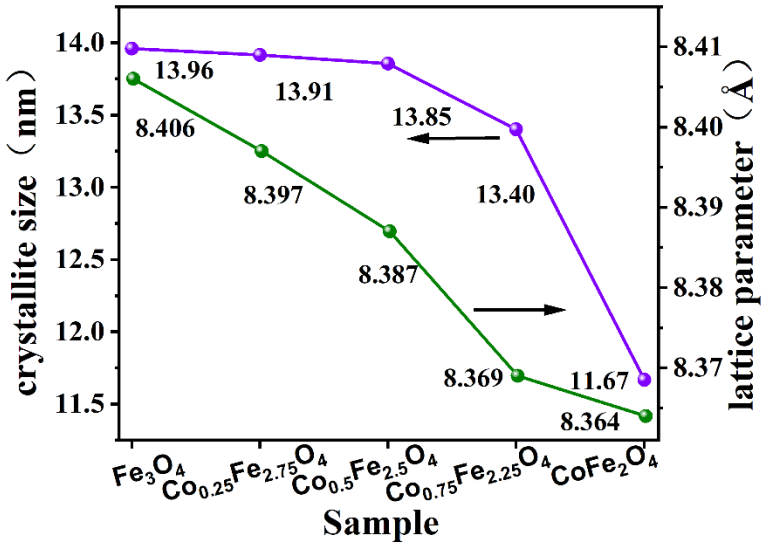


Figure S2. Variation of lattice parameter and crystallite size of $Co_xFe_{3-x}O_4$ nanoparticles.

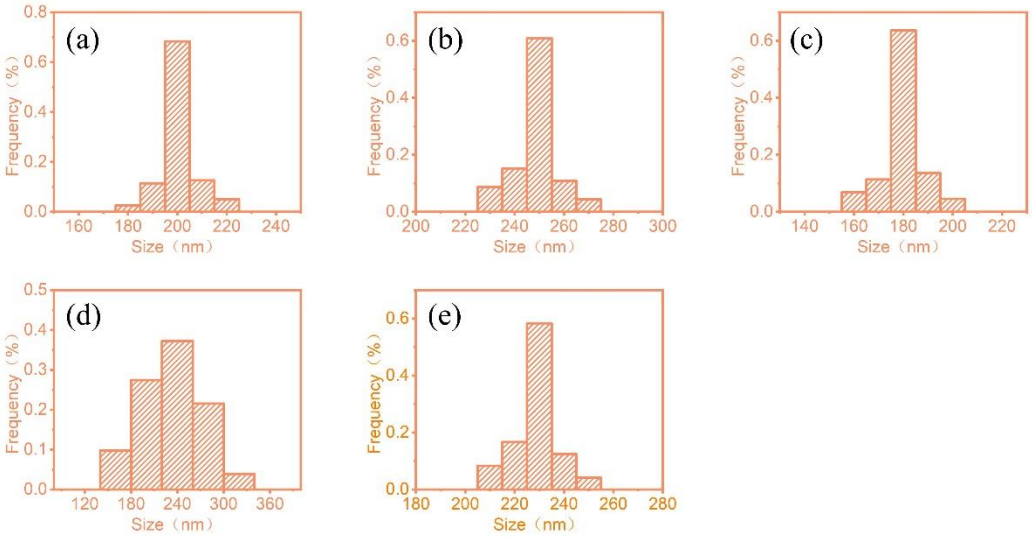


Figure S3. Size distribution of $Co_xFe_{3-x}O_4$ hollow nanospheres.

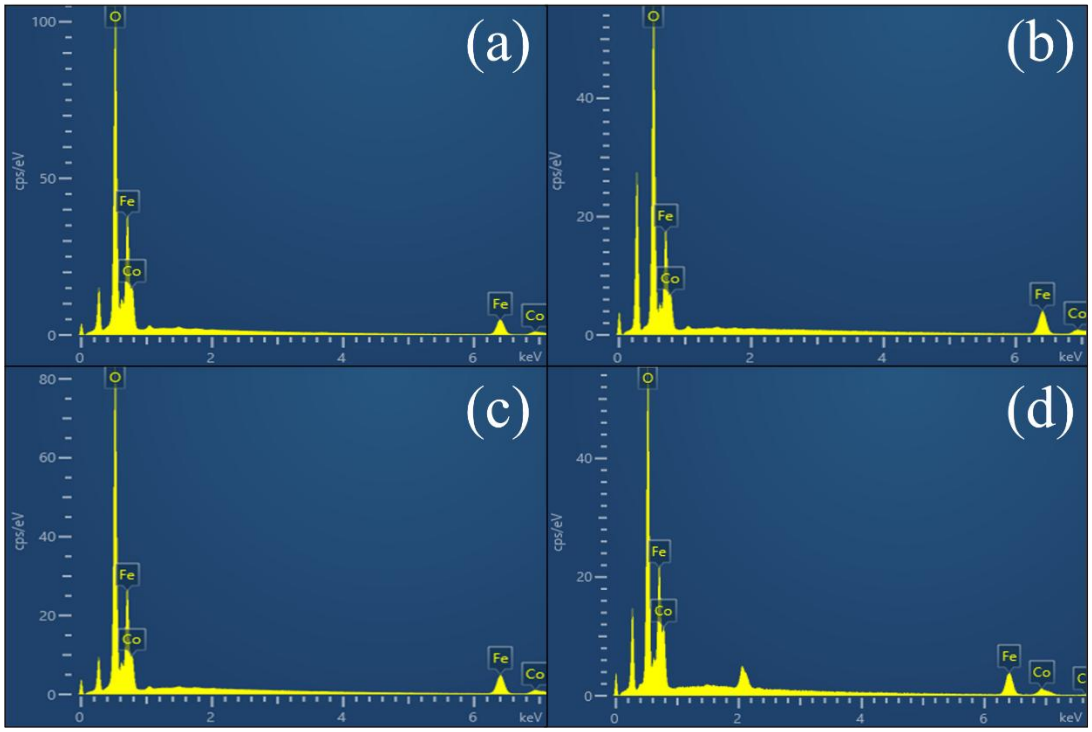


Figure S4. EDS spectra of the $Co_xFe_{3-x}O_4$ hollow nanospheres with different Co substitution: (a) x = 0.25, (b)x = 0.5, (c) x = 0.75, (d) x = 1.

Table S1. The table that summarizes the measured compositions.

Sample Name $Co_xFe_{3-x}O_4$	Atomic%		Actual Composition
	Co	Fe	
x = 0	0	1	$Fe_{2.861}O_4$
x = 0.25	0.081	0.919	$Co_{0.243}Fe_{2.757}O_4$
x = 0.5	0.189	0.811	$Co_{0.567}Fe_{2.433}O_4$
x = 0.75	0.244	0.756	$Co_{0.732}Fe_{2.268}O_4$
x = 1	0.306	0.694	$Co_{0.918}Fe_{2.082}O_4$

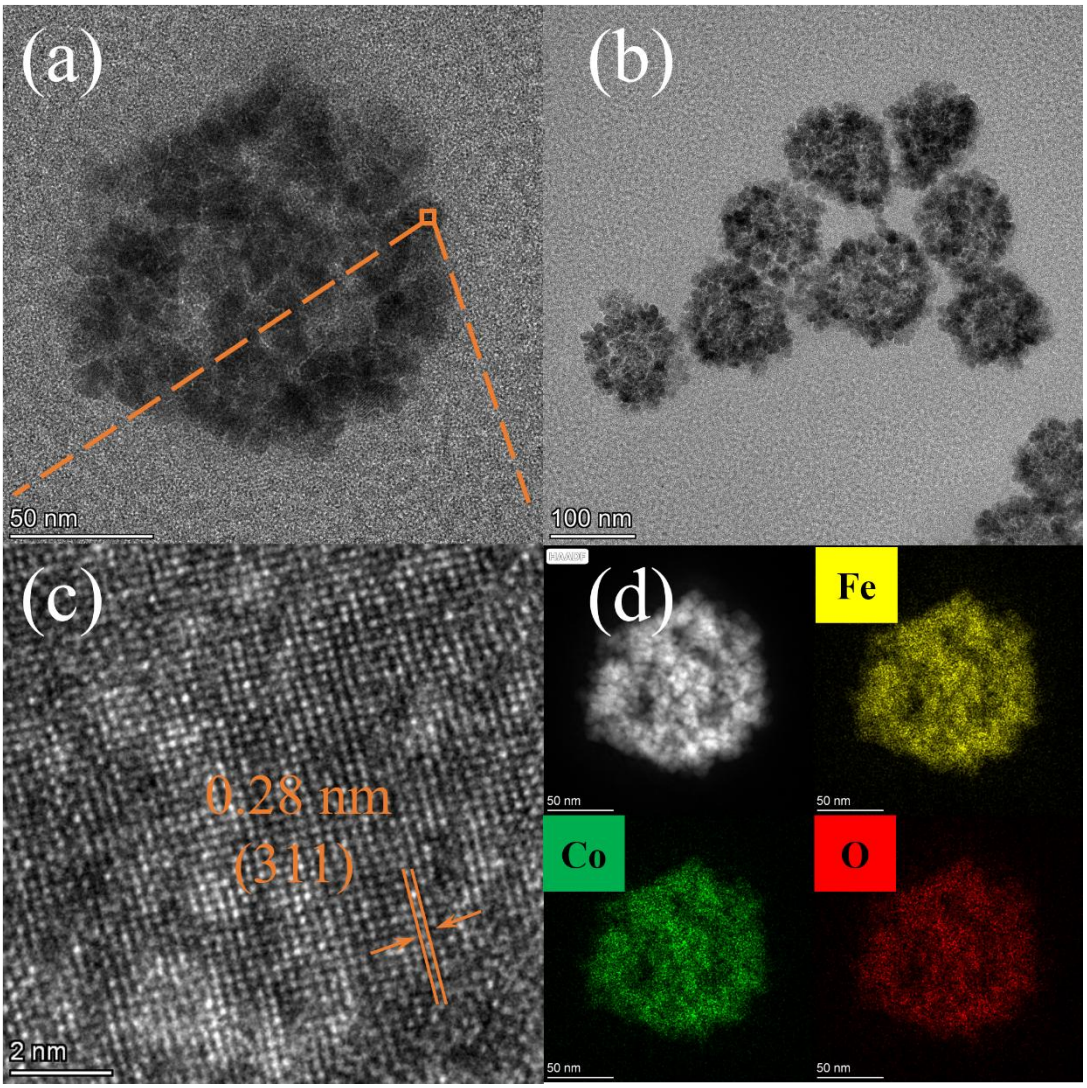


Figure S5. (a,b) TEM image and (c) HRTEM image of $Co_{0.75}Fe_{2.25}O_4$ hollow nanospheres; corresponding elemental mapping images of (d) $Co_{0.75}Fe_{2.25}O_4$ hollow nanospheres.

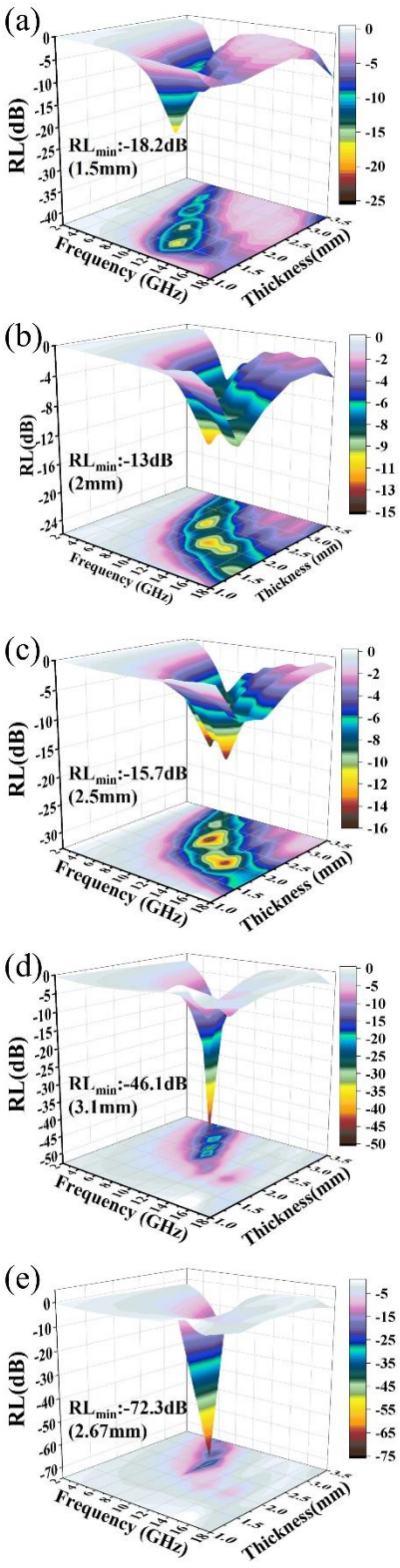


Figure S6. Three-dimensional results of the RL performances for the five samples. (a) Fe_3O_4, (b) $Co_{0.25}Fe_{2.75}O_4$, (c) $Co_{0.5}Fe_{2.5}O_4$, (d) $Co_{0.75}Fe_{2.25}O_4$ and (e) $CoFe_2O_4$.

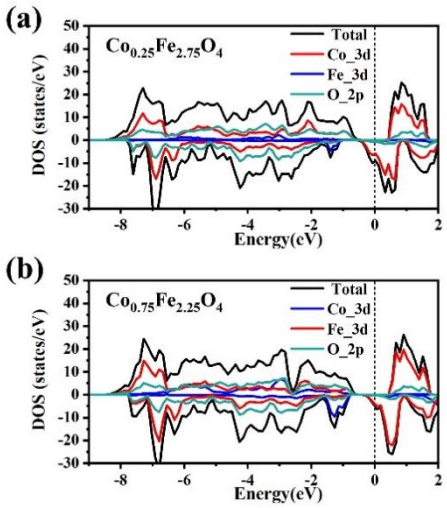


Figure S7. Density of states (DOS) of (a,) $Co_{0.25}Fe_{2.75}O_4$, (b) $Co_{0.75}Fe_{2.25}O_4$. The Fermi level is set to zero energy and indicated by the vertical dotted line.

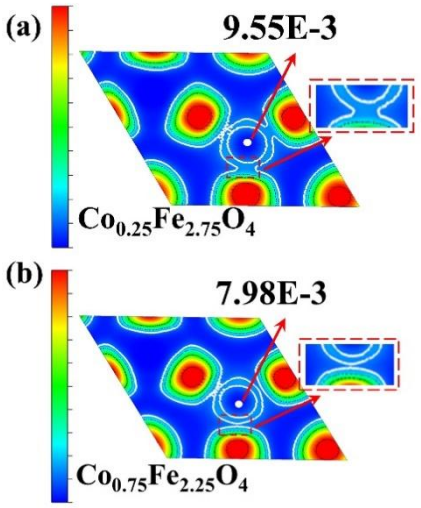


Figure S8. Electron localization function (ELF) crystal surface (100) of (a,) $Co_{0.25}Fe_{2.75}O_4$, (b) $Co_{0.75}Fe_{2.25}O_4$ (from blue to red indicates a gradual strengthening of electron localization).

Table S2. The magnetic moments of each atom and the total magnetic moment of each molecular formula in $Co_xFe_{3-x}O_4$ (x =0, 0.25, 0.5, 0.75, 1).

x	Fe_Oct/μ_B	Fe_Tet/μ_B	Co_Oct/μ_B	M_{total}/μ_B
x = 0	-3.828	3.948	---	4.00
x = 0.25	-3.872	3.921	2.397	3.75
x = 0.5	-3.916	3.931	2.335	3.50
x = 0.75	-4.004	4.052	2.588	3.25
x = 1	-3.932	4.108	2.524	3.00

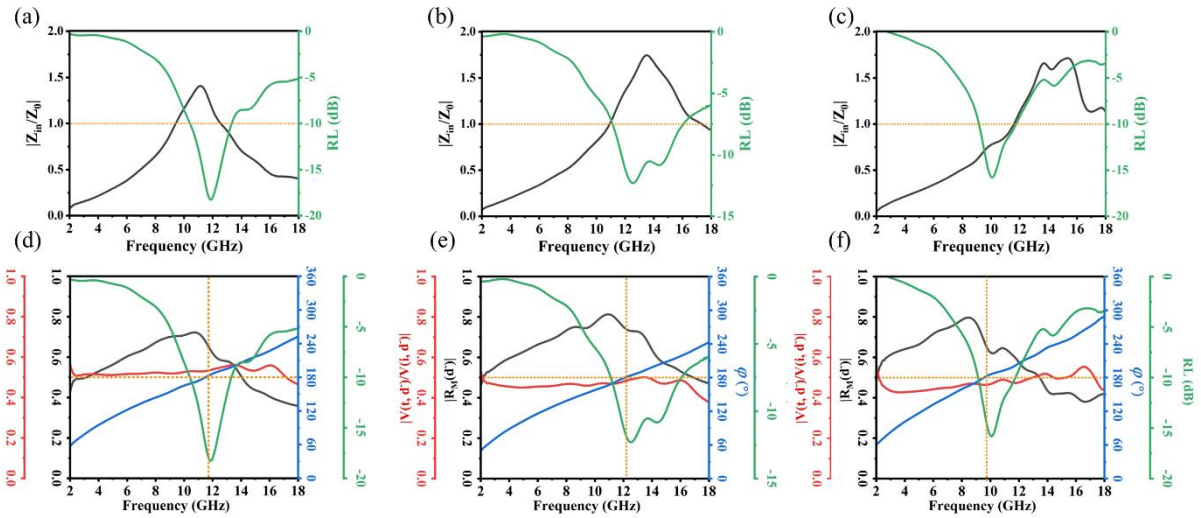


Figure S9. RL_{min}(dB) curve and $|Z_{in}/Z_0|$ values of (a) Fe_3O_4, (b) $Co_{0.25}Fe_{2.75}O_4$ and (c) $Co_{0.5}Fe_{2.5}O_4$; the RL_{min}(dB) curves, phase difference curves, $R_M(d^-)$ curves, and $|V(t,d^-)/V(i,d^-)|$ curves of (d) Fe_3O_4, (e) $Co_{0.25}Fe_{2.75}O_4$ and (f) $Co_{0.5}Fe_{2.5}O_4$.

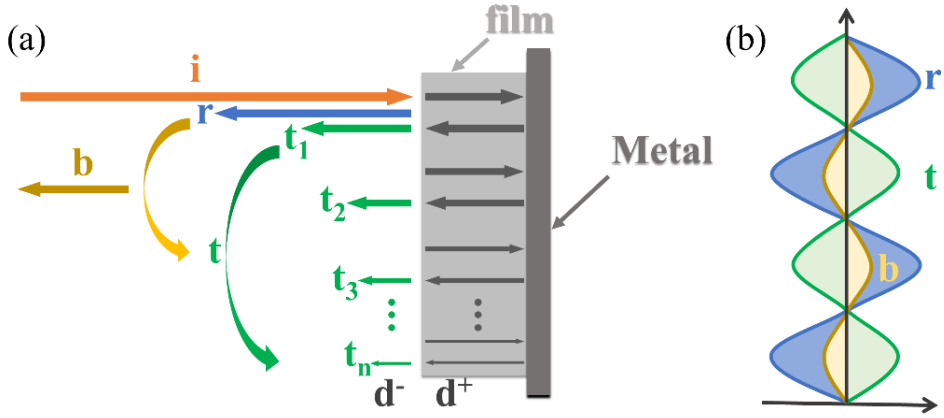


Figure S10(a, b). The diagram of wave cancellation theory.

Table S3. Calculation results of wave cancellation theory for $Co_xFe_{3-x}O_4$.

x	f/(GHz)	ε'	ε''	μ'	μ''	d/(mm)	$R_M(d^-)$	$V(t,d^-)/V(i,d^-)$	Phase-difference/(°)
1	9.2	10.67	0.97	0.83	0.24	2.67	$0.5587e^{-3.080i}$	$0.5589e^{0.059i}$	179.85
0.75	8.4	10.50	1.10	0.76	0.19	3.10	$0.5724e^{-3.090i}$	$0.0.5696e^{0.051i}$	179.96
0.5	10.08	8.42	2.16	1.14	0.08	2.50	$0.4696e^{3.062i}$	$0.6211\ e^{-0.125i}$	182.60
0.25	12.88	8.94	1.77	1.04	0.06	2.00	$0.4958e^{3.075i}$	$0.7289e^{-0.108i}$	182.37
0	11.84	12.05	2.34	1.12	0.13	1.50	$0.5355e^{3.062i}$	$0.6368e^{-0.113i}$	181.91

Appendix

The derivations for the formula for reflection loss

According to Fresnel equations, the reflection coefficient $R_M(d^-)$ is as follows:

$$R_M(d^-) = \frac{V(r,d^-)}{V(i,d^-)} = \frac{Z_M - Z_0}{Z_M + Z_0} = \left|\frac{Z_M - Z_0}{Z_M + Z_0}\right| e^{j\varphi_1} \quad (1)$$

where φ_1 is the phase of $R_M(d^-)$, $V(r, d^-)$ and $V(i, d^-)$ are the voltages at the interface, and Z_M and Z_0 are the characteristic impedances of a material and free space, respectively. Z_M is expressed as follows:

$$Z_M = \sqrt{\frac{\mu_0 \mu_r}{\varepsilon_0 \varepsilon_r}} = Z_0 \sqrt{\frac{\mu_r}{\varepsilon_r}} \quad (2)$$

From Eq. (1), $R_M(d^-)$ is only related to Z_M and Z_0, which are characteristic properties of the film. That is the amplitude and phase of the reflected beam r depend only on the intrinsic impedance Z_M of the material.

Based on the WCT, the voltage ratio of the total transmitted beam t to the incident beam i $\left(\frac{V(t,d^-)}{V(i,d^-)}\right)$ could be obtained by means of electromagnetic boundary conditions:

$$\frac{V(t,d^-)}{V(i,d^-)} = \frac{1}{V(i,d^-)}[V(t_1,d^-) + V(t_2,d^-) + V(t_3,d^-) + \cdots + V(t_n,d^-)]$$
$$= \frac{-[1-R_M^2(d^-)]e^{-2\gamma d}}{1-R_M(d^-)e^{-2\gamma d}} \quad (3)$$

in which d is the thickness of the film device, γ is the propagation constant of the electromagnetic wave in the film filled with absorber, $\gamma = j\frac{2\pi f}{c}\sqrt{\mu_r \varepsilon_r} = \alpha + j\beta$, α represents the attenuation coefficient and β the phase constant [1, 2]. From Eq. (3), the phase of the transmitted beam t depends on d. From Eqs. (1) and (3), the expression for RL can be obtained as:

$$RL = \frac{V(b,d^-)}{V(i,d^-)} = \frac{V(t,d^-)}{V(i,d^-)} + \frac{V(r,d^-)}{V(i,d^-)} = R_M(d^-) + \frac{-[1-R_M^2(d^-)]e^{-2\gamma d}}{1-R_M(d^-)e^{-2\gamma d}}$$
$$= \frac{R_M(d^-) - e^{-2\gamma d}}{1-R_M(d^-)e^{-2\gamma d}}$$
$$= \frac{Z_{in}(d^-) - Z_0}{Z_{in}(d^-) + Z_0} \quad 4)$$

where the input impedance (Z_{in}) based on the metal backing model is [3]:

$$Z_{in}(d^-) = Z_M \tanh(\gamma d) = Z_0 \sqrt{\frac{\mu_r}{\varepsilon_r}} \tanh[j\frac{2\pi f d}{c}\sqrt{\mu_r \varepsilon_r}] \quad (5)$$

Then, the RL in decibels (dB) can also be written as

$$RL(dB) = 20lg\left|\frac{Z_{in}(d^-) - Z_0}{Z_{in}(d^-) + Z_0}\right| \quad (6)$$

It is worth noting that in Eqs. (1) - (4), the phase effect of each beam is considered. In summary, RL is evaluated by the reflected beam r and transmitted beam t at d^- which has been shown to be in the WCT. RL_{min} is accurately related to the amplitude and phase

of beams *r* and *i* [4].

References

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